

Research paper

Prediction of solvation free energy and partition coefficient of polychlorinated biphenyls using thermodynamic integration

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HIGHLIGHTS

- The quick prediction of the solvation free energy and partition coefficient of PCBs was calculated using thermodynamic integration method.
- The comparison of two methods of integrations for computation accuracy of solvation free energy.
- The partial charges of PCBs were optimized for improving the accuracy of prediction.

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ABSTRACT

Partition coefficient of Polychlorinated biphenyls (PCBs) is important property for environment governance. We employed the all-atom molecular dynamics (MD) simulation to calculate solvation free energies of the PCBs in water and 1-octanol, respectively. The solvation free energy was predicted using thermodynamics integration (TI) approach. Two methods were compared to calculate the integrations of free energy in the TI approach. To improve the accuracy of our calculations, optimized atomic partial charges were applied for the PCBs in our MD simulations, leading to good computational predictions in partition coefficients. Our computational studies provided reliable data for separation process design.

1. Introduction

Polychlorinated biphenyls (PCBs) that are important chemical products have been used as dielectric and coolant fluids in transformers, capacitors, and electric motors [1]. However, it is reason that the PCBs' toxicity and classification as a persistent organic pollutant, so PCBs production was banned by United States Congress in 1979 and by the Stockholm Convention on Persistent Organic Pollutants in 2001 [1]. Hundreds of million pounds of the PCBs pollutants have been released into the environment. The PCBs are existed in the earth and water for the pollution by industry production process, so their physical chemistry properties would help to environments governance. Currently, around 150 out of a total of 209 PCB congeners can be found in our living environment.

Partition coefficient of the PCBs is a key technical indicator for scientists to assess the quality of air in environment. There are a lot of papers reporting the properties of PCBs including partition coefficient, but the collection of these information is not cover every compound of

PCBs, it is reason that the accurate determination of all PCB congeners may be difficult and expensive in terms of cost and time, or even some of which might not have synthesized or purified. A number of computational methods were developed to estimate the partition coefficient. Some molecular descriptors were parameterized for different chlorine atoms sites, as Han et al. [2] studies using quantitative structure-activity relationship (QSAR) method. In their work, two same type estimated equations were given, their three N variables only relating with numbers of the chlorine atoms site, so the detail structural information of real molecular structures was ignored. The advantage of this model allows us to quickly calculate the partition coefficient. Abraham proposed a new method to calculate solvation free energy, and in their model, the free energies were separated into five different contributions which are mapping to the molecular properties. Abraham method [3,4] was widely applied in environment science because it contains complete parameters and accurate estimated values for the PCBs. Some researchers utilized quantum mechanics calculation to evaluate solvation free energy, as COSMO-SAC model [5] studying by Philips et al.

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works, this model uses the molecular quantum COSMO data to predict partition coefficient for any kinds of chemical compounds. It is valuable to extending its applications as prediction method for the COSMO-SAC model. van der Spoel et al. still employed several different quantum methods as HF, BP86, COSMO-RS, XlogP3, KowWin and GAFF-ESP to calculate partition coefficient of PCBs [6]. And molecular dynamics (MD) simulation is an effective method, which can investigate the molecular trajectory and free energy without experiments. It could obtain accurate results once the appropriate force fields and atomic charges were chosen.

In this work, we utilized the molecular dynamic simulation from the open-source codes to study the solvation free energy and partition coefficient of PCB congeners. The solvation free energy was calculated using thermodynamic integration (TI) approach [7–10]. The similar work was studied using united-atoms force field by Garrido et al. [11], but they only reported six PCB congener results. We chose 36 PCB congeners as our objects and all-atoms force field was applied in our MD simulations. The optimized partial charge of PCBs was supported in this work.

2. Molecular models and simulation methods

2.1. Thermodynamic Integration (TI) approach

All MD simulations were carried out in GROMACS 4.5.3 software [12,13]. Thermodynamic integration (TI) method was proposed by Krikwood [8,10,14] and is widely used for computing solvation free energy of complex solutes [15]. The change of free energy related to a transition process between two generic well-defined states is studied by the TI approach. Free energy was described by the Hamiltonians from an initial reference state (state = 0) H_0 to a final target state (state = 1) H_1 . A coupling parameter λ is added in the Hamiltonian H to describe the transition between the two states $H_0 \rightarrow H_1$. Later, the integration along a continuous path of the derivatives of the free energy from the initial and final states is calculated to obtain the energy differences between the two states [16,17]. The free energy calculation for thermodynamic integration method follows as:

$$\Delta G = \int_0^1 \left\langle \frac{\partial H(\mathbf{p}, \mathbf{q}, \lambda)}{\partial \lambda} \right\rangle_{\lambda} d\lambda \quad (1)$$

where the angular brackets indicate an ensemble average at a particular λ . The ΔG is the difference of energy between the two states (state 0 $\rightarrow$ state 1). The $\mathbf{p}$ and $\mathbf{q}$ are the linear momentum and the atomic position, respectively. In fact, the free energy is calculated by setting several discrete and independent values of λ between 0 and 1, equilibrium averages can be used to evaluate derivatives of the free energy with respect to λ [10,14]. Thus, Eq. (1) can be rewritten as:

$$\Delta G = \sum_{i=1}^n \left\langle \frac{\partial H(\mathbf{p}, \mathbf{q}, \lambda_i)}{\partial \lambda_i} \right\rangle_{\lambda_i} \Delta \lambda \quad (2)$$

where the λ_i is the value between 0 and 1, and n is the total numbers of independent λ_i . The $\Delta \lambda$ is a difference between λ_i and λ_{i+1} . A deviation of calculations exists when we evaluate the free energy with Eqs. (1) and (2); two reasons lead to it: (1) the statistical error in the ensemble average of the Hamiltonian derivative at each λ_i , and (2) the error associated with the integration along the profile of free energy. The first error, in principle, will be reduced by increasing the simulation time. The second error will be decreased as long as we have sufficient intermediate points. Indeed, it has been concluded that the precision of the TI methodology depends mostly on the smoothness of the $\partial H / \partial \lambda$ vs λ plot. Theoretically, if the n of Eq. (2) is increased to infinite, the result of calculation in Eq. (2) is as accurate as that in Eq. (1). However, a large n means that more independent simulations have to be run in our finite computational resources. Thus, a good integration method that is chosen to evaluate the free energy can greatly save our computing time

in this work.

Two integration methods were applied in Eq. (2) to evaluate the free energy. The first one is the trapezoidal method [17], which is given as follows:

$$\int_{x_1}^{x_N} f(x) dx = \sum_{i=1}^{N-1} (x_{i+1} - x_i) \frac{f(x_{i+1}) + f(x_i)}{2} \quad (3)$$

where N is the total number of points and $f(x)$ is an integration function.

The second one was proposed by Jorge et al. [17] who suggest regressing the simulation data using appropriate mathematical expression. The expression is expected to describe the integration function $f(x)$ in Eq. (3). In our works, the free energy calculation was separated into Lennard Jones (LJ) interaction and Coulombic interaction. The LJ interaction was calculated from non-bonded interaction of the solute ($\lambda_1 = 0$) to full LJ interaction of solute ($\lambda_1 = 1$) with 16 independent simulations at $\lambda_1 = [0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 1]$. The Coulombic interaction was calculated from only the LJ interaction of solute ($\lambda_2 = 0$) to full (LJ + Coulombic) interaction of the solute ($\lambda_2 = 1$) with 11 independent simulations at $\lambda_2 = [0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1]$. The simulation data associated with the changes of only LJ interaction was fitted using the equation below [11,17],

$$\left(\frac{\partial H}{\partial \lambda} \right)_{\text{LJ}} = A_0 \lambda^2 + A_1 \lambda - \frac{A_2}{\lambda^2 - A_3 \lambda + A_4} + \frac{A_2}{A_4} \quad (4)$$

where the A_i is the fitting parameters. The integration equation of Eq. (4) is given as:

$$U = 4A_4 - A_3^2 > 0$$

$$\Delta G_{\text{LJ}} = \frac{A_2}{A_4} + \frac{A_0}{3} + \frac{A_1}{2} + \frac{2A_2}{\sqrt{U}} \left[\arctan\left(-\frac{A_3}{\sqrt{U}}\right) + \arctan\left(\frac{A_3-2}{\sqrt{U}}\right) \right] \quad (5)$$

The simulation data associated with the changes of only Coulombic interaction were fitted using polynomials formula [17]

$$\left(\frac{\partial H}{\partial \lambda} \right)_{\text{elec}} = \sum_{i=1}^4 a_i \lambda^i \quad (6)$$

where the a_i are the fitting parameters. The integration equation of Eq. (6) is given as:

$$\Delta G_{\text{elec}} = \frac{a_1}{2} + \frac{a_2}{3} + \frac{a_3}{4} + \frac{a_4}{5} \quad (7)$$

Summing the Eqs. (5) and (7), we can obtain the total solvation free energy ΔG_{total}

$$\Delta G_{\text{total}} = \Delta G_{\text{LJ}} + \Delta G_{\text{elec}} \quad (8)$$

Two methods were both used to calculate the free energy and compare their differences between them in this work. The detail analysis is discussed after chapter.

2.2. Molecular models and simulation method

All-atom MD simulations with the OPLS-AA force field [18,19] were carried out in this work to examine the solvation behaviors of the PCBs in water and 1-octanol solvents. The initial structure of 1-octanol is generated as linear molecule. The initial structure of PCBs is generated as connected two benzene rings at a same plane. The non-bonded interactions were described by the OPLS-AA force field in Table S1 (supporting information), the partial charges were optimized in this work. Details of simulation results with and without the optimized partial charges are discussed later in the text. The non-bonded interaction or van der Waals interaction of the atoms was calculated using the LJ potential [18,19]

$$V_{ij} = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \quad (9)$$

where the V_{ij} is the total potential energy, the first term in Eq. (9) is the LJ potential, the second term is the Coulombic interaction. In this study, the Coulombic interaction was evaluated using the PME (fast Particle-Mesh Ewald) method in GROMACS package. Its direct space is similar to the Ewald sum while the reciprocal part is performed with FFTs (Faster Fourier Transforms). The calculation of the LJ potential is corrected using Switch algorithm. A spherical cutoff distance of 12 Å was applied in our simulations to evaluate van der Waals and Coulombic interactions. There are three types of intra-molecular interactions: bond energy, angular energy and dihedral energy [18,19]

$$V_{\text{bond}} = \frac{1}{2} k_b (r - r_0)^2 \quad (10)$$

$$V_{\text{angle}} = \frac{1}{2} k_\theta (\theta - \theta_0)^2 \quad (11)$$

$$V_{\text{dihedral}} = \sum_{n=0}^5 C_n (\cos(\phi))^n \quad (12)$$

$$V_{\text{intra}} = V_{\text{bond}} + V_{\text{angle}} + V_{\text{dihedral}} \quad (13)$$

The total intra-molecular potential energy is sum of the Eqs. (10)–(12). The relevant parameters come from the OPLS-AA force field. The water molecule was described by SPC/E model [20] that treats water as a rigid triangle of sites. The length of the hydrogen-oxygen and hydrogen-hydrogen bonds are kept fixed at 1.0 Å and 1.633 Å [20], respectively. The SETTLE algorithm [12,13] was used to constraint the motions of water molecules in our MD simulations.

Our simulation system contains one PCB molecule and either of 500 water or 200 1-octanol molecules. For this reason, the solvation free energy of the PCB compound in the two solvents is called infinite dilution free energy. The infinite dilution free energy is the only variable that can be calculated accurately as long as the proper force field was used in the simulations. We conducted the MD simulations at two steps: the first one is to gradually increase the values of λ_1 , which leads to the increases of the van der Waals interaction, the second step is to gradually increase the values of λ_2 , which causes the increases of the electrostatic interaction. After the two steps in the MD simulations, the solute PCB molecule becomes fully solvated in the solvent molecules.

A big simulation box was built at initial state, making the density of solution lower than its actual density in real life. All the MD simulations were carried out in 5 ns in NPT ensemble to ensure that the system reaches to an equilibrated state. The temperature was controlled at 298 K using the Langevin stochastic dynamic method, and the Berendsen barostat method was employed to stabilize pressure at 1 atm [12,13]. The density of simulation box was sufficient to adjust to desired values at defined pressure and temperature, so the final state of simulation box was a new initial state for next process computation of free energy. As above described, the calculation of the free energy was separated into two parts: one is the LJ interaction and one is the Coulombic interaction. The 16 independent simulations were carried out from none interaction ($\lambda_1 = 0$) to full LJ interaction ($\lambda_1 = 1$) during the LJ contribution computation. The values of λ are listed in TI method description. To avoid the simulation crash when the solute molecule has weak interaction with solvent molecules, the soft-core algorithm [12,13] controlled the interaction of solute,

$$V_{\text{sc}} = \lambda_1 V [(\alpha \sigma^6 (1 - \lambda_1) + r^6)^{1/6}] \quad (14)$$

where the V is the normal “hard-core” pair potential as van der Waals (LJ) potentials for molecular interaction, and $\alpha = 0.5$ for the soft-core parameter [16,17], and $\sigma = 0.3$ as the radius of interaction [16,17]. There is a little modification of soft-core algorithm in GROMACS: the program variable “GMX_SCSIGMA_MIN” is set to equal 0 in Linux Bourne shell (command line as the “export GMX_SCSIGMA_MIN = 0”). The contributions of the Coulombic interaction were calculated in the 11 independent simulations from no charges of atoms ($\lambda_2 = 0$) to real charges of atoms ($\lambda_2 = 1$). It is worth noting that the LJ

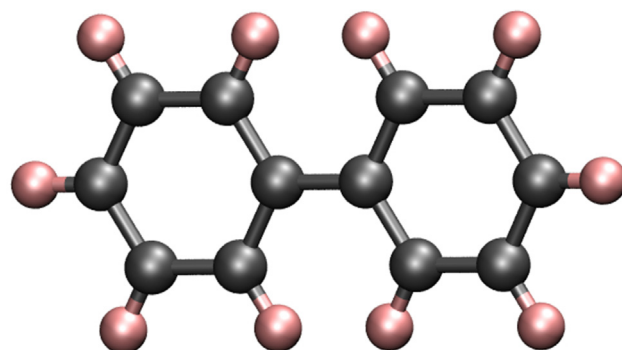


Fig. 1. The geometry structure for PCBs molecule, the black ball is carbon atoms rings, the pink ball is hydrogen or chlorine atom. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

interaction of solute was kept on at this step.

3. The results and discussion

3.1. Solvation free energy and partition coefficient computation of PCBs

The molecular structure of the PCBs is shown in Fig. 1, where all the carbon atoms within the benzene rings are in the same plane and the two benzene rings connects with a single bond. In our simulations, the molecular model of the PCBs is described as this feature controlled by dihedral angle parameters. The pink points are hydrogen or chlorine atom which is defined by real molecular structure. The LJ interactions were calculated using the OPLS force field. The partial charges of the atoms are changed accordingly as the molecular structures vary. The partial charges of special atom are treated as contribution for molecular functional group (C-H₃, C-H₂, C-H, C-O-H and C-Cl) in the OPLS force field [18,19], and the carbon atom is used to balance the partial charges of these atoms. The total charge of each PCB molecule is zero. The partial charges of chlorobenzene were used to calculate the Coulombic energy.

The solvation free energy of the PCBs in both the water and 1-octanol was divided into two parts as described above: the LJ contribution and the Coulombic contribution. Figs. 2 and 3 show the simulated results of the solvation free energy of 2-chlorobiphenyl in water and 1-octanol, respectively. In these figures, the square points are simulation

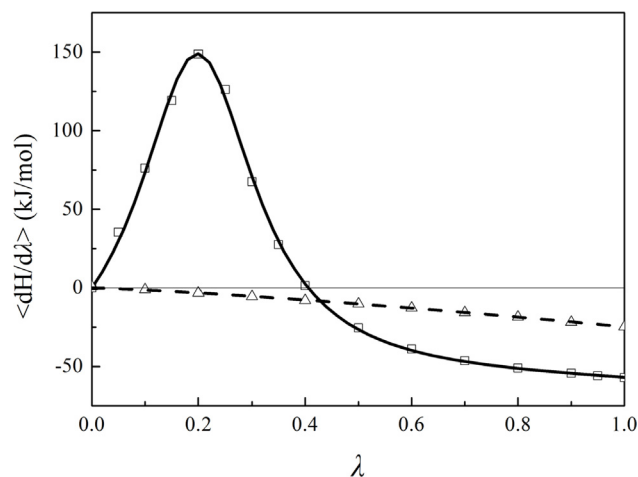


Fig. 2. The solvation free energy of 2-chlorobiphenyl in water, the points are simulation results and the lines are fitting equation results; the symbol of LJ free energy is square point and solid line, the symbol of Coulombic free energy is triangle point and dash line.

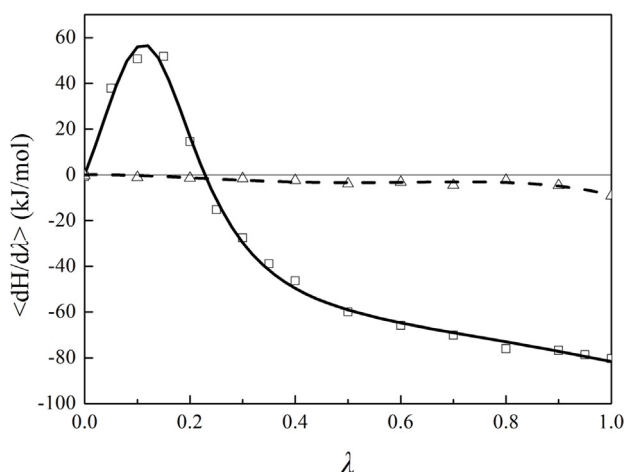


Fig. 3. The solvation free energy of 2-chlorobiphenyl in 1-octanol, the points are simulation results and the lines are fitting equation results; the symbol of LJ free energy is square point and solid line, the symbol of Coulombic free energy is triangle point and dash line.

results from the LJ contribution, and the triangle points are simulation results from the Coulombic contribution. The solid line and dash line are fitting the simulation data of solvation free energy in water and 1-octanol using Eqs. (4) and (6), the lines match well with the simulation

data in Figs. 2 and 3. It is concluded that the fitting method exhibits an accurate prediction on the calculation of the integration of solvation free energy, since these equations are analytical expressions that are easier to calculate the solvation free energy. The solvation free energy of others PCB components were calculated as the same way as that of 2-chlorobiphenyl. Further, the solvation free energy can be calculated through computing the area between fitting lines and zero line using the trapezoidal integration method.

In this work, we studied 36 PCB compounds. The solvation free energy of the PCBs in water and 1-octanol solvents was calculated using the data fitting integration method and the trapezoidal integration method, respectively. The contributions of the LJ interaction and the Coulombic interaction as well as their total Gibbs free energies were listed in Tables 1 and 2, respectively. It is worth noting that all the MD simulations were carried out using the OPLS standard charges $H = 0.115e$ and $Cl = -0.18e$. The comparison of LJ and Coulombic contributions on solvation free energies of compounds is found that the Coulombic contribution is larger than LJ contribution in water, and conversely the LJ contribution have the major contribution in solvation free energy in 1-octanol. By comparing the values in Tables 1 and 2, we clearly see that the computed solvation free energies are almost identical, indicating that both the integration methods are tailored to the calculation of solvation free energy. We believed that the fitting equations integration method could be given more reasonable values, because it is integrated by analytical equations comparing with trapezoidal method. For brevity, we will emphases on the data fitting

Table 1

The solvation free energy of PCBs in water and 1-octanol was calculated using fitting equations method by simulation using OPLS charges ($H = 0.115e$, $Cl = -0.18e$).

Compound	Solvation free energy (kJ/mol)					
	in water solution			in 1-octanol solution		
	LJ	Coul	Total	LJ	Coul	Total
2-Chlorobiphenyl	5.22	-10.88	-5.65	-38.52	-2.86	-41.38
3-Chlorobiphenyl	5.03	-10.88	-5.85	-39.11	-3.02	-42.12
4-Chlorobiphenyl	5.12	-11.34	-6.22	-40.22	-2.98	-43.19
2,2'-Dichlorobiphenyl	5.05	-10.92	-5.88	-41.58	-4.40	-45.98
2,4-Dichlorobiphenyl	4.65	-11.20	-6.56	-44.50	-3.24	-47.74
2,6-Dichlorobiphenyl	4.49	-12.26	-7.77	-42.77	-3.38	-46.15
3,3'-Dichlorobiphenyl	4.69	-9.33	-4.64	-45.33	-3.37	-48.69
3,5-Dichlorobiphenyl	4.46	-10.75	-6.29	-45.10	-2.15	-47.25
4,4'-Dichlorobiphenyl	4.52	-9.94	-5.42	-43.89	-3.53	-47.42
2,2',6-Trichlorobiphenyl	5.60	-10.76	-5.16	-44.89	-2.88	-47.78
2,3,4-Trichlorobiphenyl	3.23	-16.14	-12.91	-50.47	-5.73	-56.20
2,4,5-Trichlorobiphenyl	3.87	-14.07	-10.20	-49.81	-4.11	-53.92
2,4,6-Trichlorobiphenyl	3.82	-13.87	-10.06	-48.89	-2.48	-51.37
3,3',5-Trichlorobiphenyl	3.59	-9.91	-6.32	-50.20	-2.48	-52.68
2,2',3,3'-Tetrachlorobiphenyl	3.41	-16.88	-13.48	-50.40	-8.60	-59.01
2,2',4,5'-Tetrachlorobiphenyl	3.80	-10.30	-6.50	-52.12	-4.90	-57.01
2,2',5,5'-Tetrachlorobiphenyl	3.79	-12.15	-8.36	-52.57	-3.78	-56.35
2,2',6,6'-Tetrachlorobiphenyl	6.26	-10.46	-4.21	-46.57	-3.45	-50.02
2,3,4,5-Tetrachlorobiphenyl	2.95	-19.84	-16.90	-54.60	-7.16	-61.75
3,3',4,4'-Tetrachlorobiphenyl	2.62	-15.82	-13.20	-56.59	-7.75	-64.34
3,3',5,5'-Tetrachlorobiphenyl	3.03	-11.69	-8.66	-56.50	-1.62	-58.12
2,2',4,5,5'-Pentachlorobiphenyl	3.56	-16.20	-12.64	-57.75	-3.63	-61.38
2,3,3',4,4'-Pentachlorobiphenyl	1.71	-18.87	-17.16	-59.03	-8.40	-67.44
2,3,4,5,6-Pentachlorobiphenyl	1.85	-26.68	-24.83	-58.72	-13.87	-72.59
2,3',4,4',5-Pentachlorobiphenyl	2.14	-17.13	-14.99	-60.89	-5.81	-66.70
2,2',3,3',4,4'-Hexachlorobiphenyl	2.14	-25.65	-23.50	-63.31	-13.20	-76.51
2,2',3,3',6,6'-Hexachlorobiphenyl	5.20	-16.16	-10.96	-56.27	-6.77	-63.03
2,2',3,4,4',5'-Hexachlorobiphenyl	2.05	-23.03	-20.98	-63.60	-10.82	-74.42
2,2',4,4',5,5'-Hexachlorobiphenyl	2.76	-20.84	-18.08	-63.45	-6.52	-69.97
2,2',4,4',6,6'-Hexachlorobiphenyl	6.15	-12.89	-6.74	-58.04	-2.50	-60.54
3,3',4,4',5,5'-Hexachlorobiphenyl	0.63	-25.06	-24.43	-65.50	-12.31	-77.80
2,2',3,4,4',5,5'-Heptachlorobiphenyl	1.92	-28.32	-26.40	-68.98	-13.01	-81.99
2,2',3,3',4,4',5,5'-Octachlorobiphenyl	0.48	-36.47	-35.98	-72.07	-17.84	-89.91
2,2',3,3',5,5',6,6'-Octachlorobiphenyl	4.00	-24.48	-20.48	-66.31	-8.30	-74.62
2,2',3,3',4,5,5',6,6'-Nonachlorobiphenyl	3.54	-32.12	-28.58	-70.31	-16.57	-86.87
Decachlorobiphenyl	1.98	-40.21	-38.22	-76.22	-23.06	-99.27

Table 2

The solvation free energy of PCBs in water and 1-octanol were calculated using trapezoidal integration method by simulation using OPLS charges (H = 0.115e, Cl = -0.18e).

Compound	Solvation free energy (kJ/mol)					
	in water solution			in 1-octanol solution		
	LJ	Coul	Total	LJ	Coul	Total
2-Chlorobiphenyl	5.68	-10.85	-5.17	-38.71	-2.96	-41.67
3-Chlorobiphenyl	5.70	-10.86	-5.17	-39.16	-2.96	-42.11
4-Chlorobiphenyl	5.73	-11.32	-5.59	-40.29	-3.02	-43.31
2,2'-Dichlorobiphenyl	5.63	-10.90	-5.28	-41.93	-4.45	-46.38
2,4-Dichlorobiphenyl	5.23	-11.21	-5.98	-44.75	-3.27	-48.02
2,6-Dichlorobiphenyl	5.18	-12.26	-7.08	-43.12	-3.41	-46.53
3,3'-Dichlorobiphenyl	5.45	-9.34	-3.89	-45.59	-3.35	-48.95
3,5-Dichlorobiphenyl	5.18	-10.75	-5.57	-45.34	-2.09	-47.43
4,4'-Dichlorobiphenyl	5.17	-9.92	-4.75	-44.14	-3.59	-47.73
2,2',6-Trichlorobiphenyl	6.27	-10.76	-4.50	-45.32	-2.93	-48.25
2,3,4-Trichlorobiphenyl	3.84	-16.16	-12.32	-50.73	-5.77	-56.50
2,4,5-Trichlorobiphenyl	4.73	-14.08	-9.35	-49.80	-4.04	-53.84
2,4,6-Trichlorobiphenyl	4.48	-13.88	-9.40	-49.13	-2.48	-51.61
3,3',5-Trichlorobiphenyl	4.19	-9.93	-5.75	-50.37	-2.49	-52.86
2,2',3,3'-Tetrachlorobiphenyl	3.96	-16.90	-12.94	-50.46	-8.77	-59.23
2,2',4,5'-Tetrachlorobiphenyl	4.47	-10.33	-5.86	-52.68	-4.92	-57.60
2,2',5,5'-Tetrachlorobiphenyl	4.42	-12.17	-7.75	-52.90	-3.81	-56.70
2,2',6,6'-Tetrachlorobiphenyl	6.84	-10.47	-3.62	-47.17	-3.45	-50.62
2,3,4,5-Tetrachlorobiphenyl	3.78	-19.87	-16.09	-54.79	-7.06	-61.85
3,3',4,4'-Tetrachlorobiphenyl	3.37	-15.86	-12.50	-56.89	-7.63	-64.52
3,3',5,5'-Tetrachlorobiphenyl	3.85	-11.73	-7.88	-56.73	-1.59	-58.32
2,2',4,5,5'-Pentachlorobiphenyl	4.34	-16.26	-11.91	-58.02	-3.60	-61.62
2,3,3',4,4'-Pentachlorobiphenyl	2.44	-18.91	-16.47	-59.38	-8.50	-67.88
2,3,4,5,6-Pentachlorobiphenyl	2.68	-26.70	-24.02	-59.07	-13.90	-72.96
2,3',4,4',5-Pentachlorobiphenyl	2.96	-17.18	-14.22	-61.17	-5.87	-67.04
2,2',3,3',4,4'-Hexachlorobiphenyl	3.13	-25.70	-22.57	-63.64	-13.21	-76.85
2,2',3,3',6,6'-Hexachlorobiphenyl	5.85	-16.19	-10.34	-56.77	-6.82	-63.59
2,2',3,4,4',5'-Hexachlorobiphenyl	2.94	-23.07	-20.14	-64.00	-10.80	-74.79
2,2',4,4',5,5'-Hexachlorobiphenyl	3.67	-20.89	-17.22	-63.81	-6.49	-70.30
2,2',4,4',6,6'-Hexachlorobiphenyl	6.75	-12.93	-6.18	-58.64	-2.42	-61.06
3,3',4,4',5,5'-Hexachlorobiphenyl	1.64	-25.14	-23.50	-65.63	-12.30	-77.93
2,2',3,4,4',5,5'-Heptachlorobiphenyl	2.80	-28.36	-25.56	-69.43	-13.07	-82.50
2,2',3,3',4,4',5,5'-Octachlorobiphenyl	1.50	-36.54	-35.04	-72.36	-17.93	-90.30
2,2',3,3',5,5',6,6'-Octachlorobiphenyl	4.86	-24.52	-19.66	-66.90	-8.29	-75.19
2,2',3,3',4,5,5',6,6'-Nonachlorobiphenyl	4.46	-32.14	-27.68	-70.92	-16.67	-87.59
Decachlorobiphenyl	2.83	-40.22	-37.39	-76.77	-22.79	-99.56

integration method for a further discussion later.

Partition coefficient describes how a solute is distributed between two immiscible solvents. In this work, the partition coefficients of the PCBs in both water and 1-octanol solvents were calculated by the Gibbs free energies of the solute PCB molecule in water phase and in octanol phase, as given in Equation (4) [8]

$$\log K_{o/w} = \frac{\Delta G_w - \Delta G_o}{2.303RT} \quad (15)$$

where the $K_{o/w}$ is the partition coefficient in water/octanol system, ΔG_w and ΔG_o is Gibbs solvation free energy changes of the PCBs in water and in 1-octanol, and R is gas constant.

In Table S2 (Supporting Information), the partition coefficients were calculated relying on the values of solvation free energy from the two integration methods in Tables 1 and 2. Our results revealed that the computed partition coefficients in the two integration methods are close to each other, however, all of them are higher than the experimentally measured data [21–25]. It means that the force field parameters used in this work needs a further improvement to describe the real state of the PCBs in water and 1-octanol solvents. As well as we know, the partial charges of atoms are changed accordingly as the molecular structures varies. However, the partial charges of atoms in the OPLS standard force field are constant, which were used in this work to calculate the Coulombic interactions between the solute and the solvents. As aforementioned, the partial charge of chlorobenzene was optimized to accurately describe the solvation behaviors of the

PCBs in solvents.

3.2. Optimization of PCBs charges

We abided by the protocol of the OPLS charge optimization method to obtain the new partial charges of the hydrogen and chlorine atoms in the PCBs. It is assumed that special atoms contain the same charges for all the PCB molecules, and the carbon atom is considered as charge balance atom for the whole group. Once the partial charges of hydrogen and chlorine atoms are determined, the atomic charges of carbon atoms are defined accordingly. The 36 PCB molecules were chosen as database for the charge optimization. Setting the difference of partition coefficients between the computed values and experimentally measured values as an objective function, we minimize the root mean square error of these values:

$$RMS = \left(\frac{\sum_i^N (\log K_{o/w,i}^{\text{exp}} - \log K_{o/w,i}^{\text{cal}})^2}{N} \right)^{1/2} \quad (16)$$

where the N is the number of basement data, and superscripts of “exp” and “cal” are denoted to experimental data and calculation results, respectively.

The 2-chlorobiphenyl, 3-chlorobiphenyl and 4-chlorobiphenyl were used to optimize the hydrogen atomic partial charges, because they

Table 3

The solvation free energy of PCBs in water and 1-octanol was calculated using fitting equations method by simulation using optimization charges ($H = 0.16e$, $Cl = -0.27e$).

Compound	Solvation free energy (kJ/mol)					
	in water solution			in 1-octanol solution		
	LJ	Coul	Total	LJ	Coul	Total
2-Chlorobiphenyl	5.22	-23.37	-18.15	-38.52	-6.51	-45.03
3-Chlorobiphenyl	5.03	-23.80	-18.76	-39.11	-5.88	-44.99
4-Chlorobiphenyl	5.12	-24.67	-19.55	-40.22	-5.88	-46.09
2,2'-Dichlorobiphenyl	5.05	-23.54	-18.50	-41.58	-9.14	-50.72
2,4-Dichlorobiphenyl	4.65	-24.91	-20.26	-44.50	-7.69	-52.19
2,6-Dichlorobiphenyl	4.49	-26.81	-22.31	-42.77	-6.71	-49.48
3,3'-Dichlorobiphenyl	4.69	-19.82	-15.13	-45.33	-7.33	-52.65
3,5-Dichlorobiphenyl	4.46	-24.02	-19.56	-45.10	-4.88	-49.98
4,4'-Dichlorobiphenyl	4.52	-21.14	-16.62	-43.89	-7.71	-51.61
2,2',6-Trichlorobiphenyl	5.60	-23.54	-17.94	-44.89	-6.92	-51.82
2,3,4-Trichlorobiphenyl	3.23	-36.19	-32.96	-50.47	-13.54	-64.02
2,4,5-Trichlorobiphenyl	3.87	-31.40	-27.53	-49.81	-8.72	-58.53
2,4,6-Trichlorobiphenyl	3.82	-30.49	-26.67	-48.89	-6.36	-55.24
3,3',5-Trichlorobiphenyl	3.59	-21.64	-18.05	-50.20	-7.05	-57.24
2,2',3,3'-Tetrachlorobiphenyl	3.41	-37.62	-34.21	-50.40	-18.16	-68.57
2,2',4,5'-Tetrachlorobiphenyl	3.80	-23.05	-19.25	-52.12	-10.62	-62.74
2,2',5,5'-Tetrachlorobiphenyl	3.79	-27.10	-23.31	-52.57	-9.39	-61.96
2,2',6,6'-Tetrachlorobiphenyl	6.26	-23.06	-16.81	-46.57	-9.10	-55.67
2,3,4,5-Tetrachlorobiphenyl	2.95	-44.53	-41.59	-54.60	-21.82	-76.41
3,3',4,4'-Tetrachlorobiphenyl	2.62	-35.20	-32.58	-56.59	-18.52	-75.11
3,3',5,5'-Tetrachlorobiphenyl	3.03	-25.82	-22.79	-56.50	-6.64	-63.14
2,2',4,5,5'-Pentachlorobiphenyl	3.56	-36.12	-32.56	-57.75	-11.20	-68.95
2,3,3',4,4'-Pentachlorobiphenyl	1.71	-42.22	-40.51	-59.03	-21.20	-80.23
2,3,4,5,6-Pentachlorobiphenyl	1.85	-59.76	-57.91	-58.72	-41.98	-100.70
2,3',4,4',5-Pentachlorobiphenyl	2.14	-38.27	-36.13	-60.89	-17.14	-78.03
2,2',3,3',4,4'-Hexachlorobiphenyl	2.14	-56.89	-54.74	-63.31	-29.48	-92.79
2,2',3,3',6,6'-Hexachlorobiphenyl	5.20	-36.01	-30.81	-56.27	-15.01	-71.27
2,2',3,4,4',5'-Hexachlorobiphenyl	2.05	-51.17	-49.12	-63.60	-26.72	-90.33
2,2',4,4',5,5'-Hexachlorobiphenyl	2.76	-46.46	-43.71	-63.45	-19.51	-82.95
2,2',4,4',6,6'-Hexachlorobiphenyl	6.15	-28.61	-22.46	-58.04	-7.19	-65.23
3,3',4,4',5,5'-Hexachlorobiphenyl	0.63	-55.49	-54.86	-65.50	-30.68	-96.18
2,2',3,4,4',5,5'-Heptachlorobiphenyl	1.92	-63.03	-61.10	-68.98	-35.80	-104.78
2,2',3,3',4,4',5,5'-Octachlorobiphenyl	0.48	-81.01	-80.53	-72.07	-48.04	-120.12
2,2',3,3',5,5',6,6'-Octachlorobiphenyl	4.00	-55.65	-51.64	-66.31	-27.44	-93.75
2,2',3,3',4,5,5',6,6'-Nonachlorobiphenyl	3.54	-73.43	-69.89	-70.31	-50.38	-120.69
Decachlorobiphenyl	1.98	-92.76	-90.78	-76.22	-68.79	-145.01

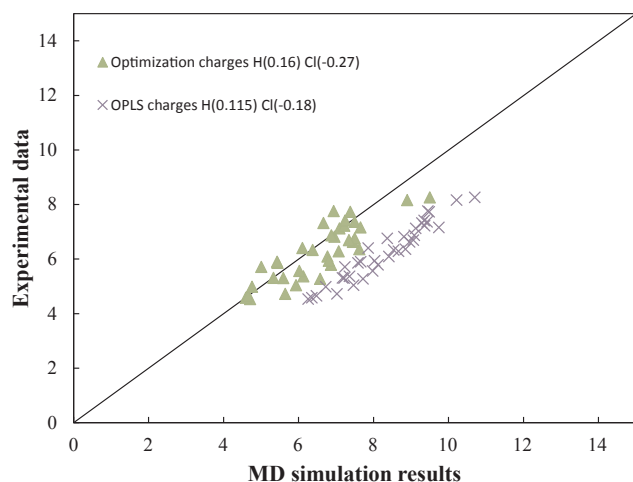


Fig. 4. The comparison between the simulation results and experiments, the triangle point is simulation results of optimization charge, and the cross point is simulation results of OPLS charge.

have only one chlorine atom. The partial charges of chlorine atom on the three PCB compounds have slight impact on the Coulombic contribution of the solvation free energy. We found that the values of hydrogen partial charges are reasonable ranging from 0.15e to 0.16e

during our optimization process. Finally, we found that the partial charges of the hydrogen and chlorine atoms are 0.16e and $-0.27e$, which have a minimum value of *RMS* in Equation (16). The *RMS* value of optimization charges is improved as 0.65 from original 2.08.

The calculated solvation free energies with the uses of the optimized partial charges were listed in Table 3, which are lower than the ones in Table 1. We employed the optimized partial charges to re-simulate the solvation behaviors of the 36 PCBs in water and 1-octanol, and re-calculated their partition coefficients in the water/octanol phases. The new results are reasonably agreed with the experimental data [21–25], as compared in Fig. 4 and Table S3 (Supporting Information).

4. Conclusions

The PCBs is serious persistent organic pollutants and their physical chemistry properties are important information for environments governance. There are many methods to estimate the PCBs properties as the solvation free energy and partition coefficient. In our work the solvation free energy and partition coefficient of PCBs was calculated using OPLS force field by molecular dynamics simulation. We found that the OPLS charges could not be describe the real state of PCBs in water/1-octanol system. The computation of partition coefficient is higher than the experiments, so we tried to optimize the partial charge for PCBs.

The *RMS* value of object function is minimization when the partial charges of hydrogen and chlorine atoms are 0.16e and $-0.27e$, respectively. The new results of partition coefficient have obvious

improvement comparing with the old ones. The conclusion is that the real state of PCBs in water/1-octanol can be described by the new partial charges for hydrogen and chlorine atoms. Our work would be support the more choices for estimated properties of PCBs compounds.

CRedit authorship contribution statement

Li Yang: Conceptualization, Validation, Investigation, Writing - original draft. **Changjun Peng:** Writing - review & editing, Supervision. **Honglai Liu:** Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cplett.2020.137708>.

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